

14.4 Contingency Table Analysis of Two Distributions

In this section and the next three sections, we deal with *measures of association* for two distributions. The situation is this: Each data point has two or more different quantities associated with it, and we want to know whether knowledge of one quantity gives us any demonstrable advantage in predicting the value of another quantity. In many cases, one variable will be an “independent” or “control” variable, and another will be a “dependent” or “measured” variable. Then, we want to know if the latter variable *is* in fact dependent on or *associated* with the former variable. If it is, we want to have some quantitative measure of the strength of the association. One often hears this loosely stated as the question of whether two variables are *correlated* or *uncorrelated*, but we will reserve those terms for a particular kind of association (linear, or at least monotonic), as discussed in §14.5 and §14.6.

Notice that, as in previous sections, the different concepts of significance and strength appear: The association between two distributions may be very significant even if that association is weak — if the quantity of data is large enough.

It is useful to distinguish among some different kinds of variables, with different categories forming a loose hierarchy.

- A variable is called *nominal* if its values are the members of some unordered set. For example, “state of residence” is a nominal variable that (in the U.S.) takes on one of 50 values; in astrophysics, “type of galaxy” is a nominal variable with the three values “spiral,” “elliptical,” and “irregular.”
- A variable is termed *ordinal* if its values are the members of a discrete, but ordered, set. Examples are grade in school, planetary order from the Sun (Mercury = 1, Venus = 2, . . .), and number of offspring. There need not be any concept of “equal metric distance” between the values of an ordinal variable, only that they be intrinsically ordered.
- We will call a variable *continuous* if its values are real numbers, as are times, distances, temperatures, etc. (Social scientists sometimes distinguish between *interval* and *ratio* continuous variables, but we do not find that distinction very compelling.)

A continuous variable can always be made into an ordinal one by binning it into ranges. If we choose to ignore the ordering of the bins, then we can turn it into a nominal variable. Nominal variables constitute the lowest type of the hierarchy, and therefore the most general. For example, a set of *several* continuous or ordinal variables can be turned, if crudely, into a single nominal variable, by coarsely binning each variable and then taking each distinct combination of bin assignments as a single nominal value. When multidimensional data are sparse, this is often the only sensible way to proceed.

The remainder of this section will deal with measures of association between *nominal* variables. For any pair of nominal variables, the data can be displayed as a *contingency table*, a table whose rows are labeled by the values of one nominal variable, whose columns are labeled by the values of the other nominal variable, and whose entries are nonnegative integers giving the number of observed events for each combination of row and column (see Figure 14.4.1). The analysis of association between nominal variables is thus called *contingency table analysis* or *cross-*

	0. red	1. green	. . .	
0. male	# of red males N_{00}	# of green males N_{01}	. . .	# of males $N_{0\cdot}$
1. female	# of red females N_{10}	# of green females N_{11}	. . .	# of females $N_{1\cdot}$
$\vdots$	$\vdots$	$\vdots$	. . .	$\vdots$
	# of red $N_{\cdot 0}$	# of green $N_{\cdot 1}$	. . .	total # N

Figure 14.4.1. Example of a contingency table for two nominal variables, here sex and color. The row and column marginals (totals) are shown. The variables are “nominal,” i.e., the order in which their values are listed is arbitrary and does not affect the result of the contingency table analysis. If the ordering of values has some intrinsic meaning, then the variables are “ordinal” or “continuous,” and correlation techniques (§14.5 – §14.6) can be utilized.

tabulation analysis.

The remainder of this section gives an approach, based on the chi-square statistic, that does a good job of characterizing the significance of association but is only so-so as a measure of the strength (principally because its numerical values have no very direct interpretations). We will return to contingency table analysis in §14.7 with an approach, based on the information-theoretic concept of *entropy*, that will say little about the significance of association (use chi-square for that!) but is capable of very elegantly characterizing the strength of an association already known to be significant.

14.4.1 Measures of Association Based on Chi-Square

Some notation first: Let N_{ij} denote the number of events that occur with the first variable x taking on its i th value and the second variable y taking on its j th value. Let N denote the total number of events, the sum of all the N_{ij} 's. Let $N_{i\cdot}$ denote the number of events for which the first variable x takes on its i th value regardless of the value of y ; $N_{\cdot j}$ is the number of events with the j th value of y regardless of x . So we have

$$\begin{aligned}
 N_{i\cdot} &= \sum_j N_{ij} & N_{\cdot j} &= \sum_i N_{ij} \\
 N &= \sum_i N_{i\cdot} = \sum_j N_{\cdot j}
 \end{aligned}
 \tag{14.4.1}$$

In other words, “dot” is a placeholder that means, “sum over the missing index”. $N_{.j}$ and $N_{i.}$ are sometimes called the *row and column totals* or *marginals*, but we will use these terms cautiously since we can never keep straight which are the rows and which are the columns!

The null hypothesis is that the two variables x and y have no association. In this case, the probability of a particular value of x given a particular value of y should be the same as the probability of that value of x regardless of y . Therefore, in the null hypothesis, the expected number for any N_{ij} , which we will denote n_{ij} , can be calculated from only the row and column totals,

$$\frac{n_{ij}}{N_{.j}} = \frac{N_{i.}}{N} \quad \text{which implies} \quad n_{ij} = \frac{N_{i.}N_{.j}}{N} \quad (14.4.2)$$

Notice that if a column or row total is zero, then the expected number for all the entries in that column or row is also zero; in that case, the never-occurring bin of x or y should simply be removed from the analysis.

The chi-square statistic is now given by equation (14.3.1), which, in the present case, is summed over all entries in the table:

$$\chi^2 = \sum_{i,j} \frac{(N_{ij} - n_{ij})^2}{n_{ij}} \quad (14.4.3)$$

The number of degrees of freedom is equal to the number of entries in the table (product of its row size and column size) minus the number of constraints that have arisen from our use of the data themselves to determine the n_{ij} . Each row total and column total is a constraint, except that this overcounts by one, since the total of the column totals and the total of the row totals both equal N , the total number of data points. Therefore, if the table is of size I by J , the number of degrees of freedom is $IJ - I - J + 1$. Equation (14.4.3), along with the chi-square probability function (§6.2), now give the significance of an association between the variables x and y . Incidentally, the two-sample chi-square test for equality of distributions, equation (14.3.3), is a special case of equation (14.4.3) with $J = 2$ and with the y variable simply a label distinguishing the two samples.

Suppose there is a significant association. How do we quantify its strength, so that (e.g.) we can compare the strength of one association with another? The idea here is to find some reparametrization of χ^2 that maps it into some convenient interval, like 0 to 1, where the result is not dependent on the quantity of data that we happen to sample, but rather depends only on the underlying population from which the data were drawn. There are several different ways of doing this. Two of the more common are called *Cramer's V* and the *contingency coefficient C*.

The formula for Cramer's V is

$$V = \sqrt{\frac{\chi^2}{N \min(I - 1, J - 1)}} \quad (14.4.4)$$

where I and J are again the numbers of rows and columns, and N is the total number of events. Cramer's V has the pleasant property that it lies between zero and one inclusive, equals zero when there is no association, and equals one only when the association is perfect: All the events in any row lie in one unique column, and vice

versa. (In chess parlance, no two rooks, placed on a nonzero table entry, can capture each other.)

In the case of $I = J = 2$, Cramer's V is also referred to as the *phi* statistic.

The contingency coefficient C is defined as

$$C = \sqrt{\frac{\chi^2}{\chi^2 + N}} \quad (14.4.5)$$

It also lies between zero and one, but (as is apparent from the formula) it can never achieve the upper limit. While it can be used to compare the strength of association of two tables with the same I and J , its upper limit depends on I and J . Therefore it can never be used to compare tables of different sizes.

The trouble with both Cramer's V and the contingency coefficient C is that, when they take on values in between their extremes, there is no very direct interpretation of what that value means. For example, you are in Las Vegas, and a friend tells you that there is a small, but significant, association between the color of a croupier's eyes and the occurrence of red and black on his roulette wheel. Cramer's V is about 0.028, your friend tells you. You know what the usual odds against you are (because of the green zero and double zero on the wheel). Is this association sufficient for you to make money? Don't ask us! For a measure of association that is directly applicable to gambling, look at §14.7.

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```
void cntab(MatInt_I &nn, Doub &chisq, Doub &df, Doub &prob, Doub &cramrv,
Doub &ccc)
```

Given a two-dimensional contingency table in the form of an array `nn[0..ni-1][0..nj-1]` of integers, this routine returns the chi-square `chisq`, the number of degrees of freedom `df`, the p -value `prob` (small values indicating a significant association), and two measures of association, Cramer's V (`cramrv`) and the contingency coefficient C (`ccc`).

```
{
    const Doub TINY=1.0e-30;           A small number.
    Gamma gam;
    Int i,j,nnj,nni,minij,ni=nn.nrows(),nj=nn.ncols();
    Doub sum=0.0,expctd,temp;
    VecDoub sumi(ni),sumj(nj);
    nni=ni;                             Number of rows...
    nnj=nj;                             ...and columns.
    for (i=0;i<ni;i++) {               Get the row totals.
        sumi[i]=0.0;
        for (j=0;j<nj;j++) {
            sumi[i] += nn[i][j];
            sum += nn[i][j];
        }
        if (sumi[i] == 0.0) --nni;       Eliminate any zero rows by reducing the number.
    }
    for (j=0;j<nj;j++) {               Get the column totals.
        sumj[j]=0.0;
        for (i=0;i<ni;i++) sumj[j] += nn[i][j];
        if (sumj[j] == 0.0) --nnj;     Eliminate any zero columns.
    }
    df=nni*nnj-nni-nnj+1;              Corrected number of degrees of freedom.
    chisq=0.0;
    for (i=0;i<ni;i++) {               Do the chi-square sum.
        for (j=0;j<nj;j++) {
            expctd=sumj[j]*sumi[i]/sum;
            temp=nn[i][j]-expctd;
            chisq += temp*temp/(expctd+TINY);
        }
    }
}
```

Here TINY guarantees that any eliminated row or column will not contribute to the sum.